



# Long-Tailed Class Incremental Learning

Hanna Hamrell - CV Reading Group @ RPL  
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## Long-Tailed Class Incremental Learning

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**Abstract.** In class incremental learning (CIL) a model must learn new classes in a sequential manner without forgetting old ones. However, conventional CIL methods consider a balanced distribution for each new

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[https://doi.org/10.1007/978-3-031-19827-4\\_29](https://doi.org/10.1007/978-3-031-19827-4_29)

[https://www.ecva.net/papers/eccv\\_2022/papers\\_ECCV/papers/136930486.pdf](https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136930486.pdf)



# Agenda

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- CIL/LT



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- LT scenarios and proposed two-stage approach to LT-CIL



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- Key takeaways



# Why you want to learn about this paper...

- Defines good LT-CIL scenarios



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- Proposes a baseline method for reducing the bias caused by long-tailed class distributions within CIL scenarios



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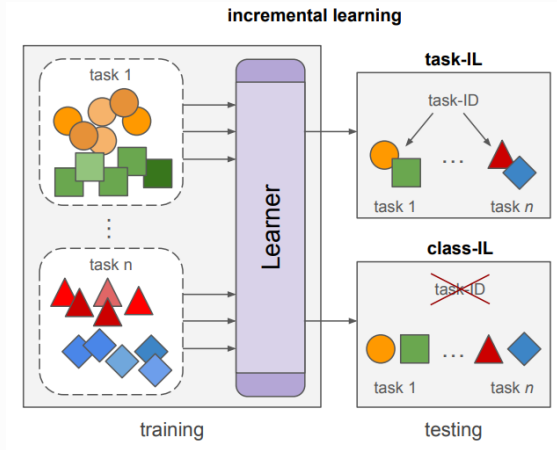
- Defines good LT-CIL scenarios
- Proposes a baseline method for reducing the bias caused by long-tailed class distributions within CIL scenarios
- Many other works in CIL limit themselves to balanced class distributions, even though long-tailed distributions are more common in the real world



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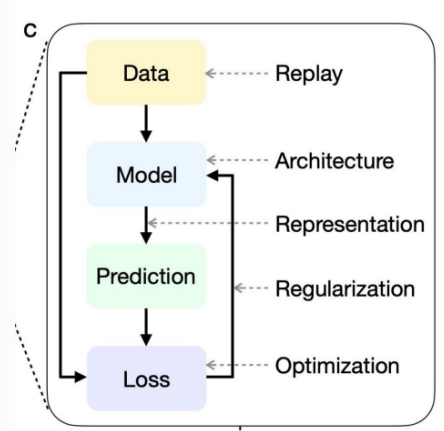
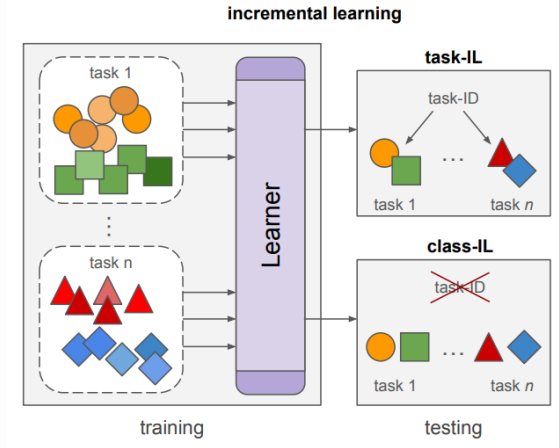
- Defines good LT-CIL scenarios
- Proposes a baseline method for reducing the bias caused by long-tailed class distributions within CIL scenarios
- Many other works in CIL limit themselves to balanced class distributions, even though long-tailed distributions are more common in the real world
- Can be applied to other CIL methods

# Class Incremental Learning



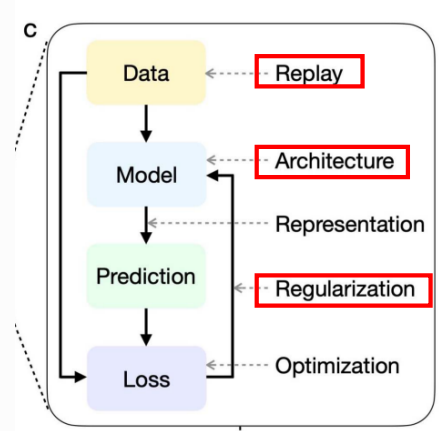
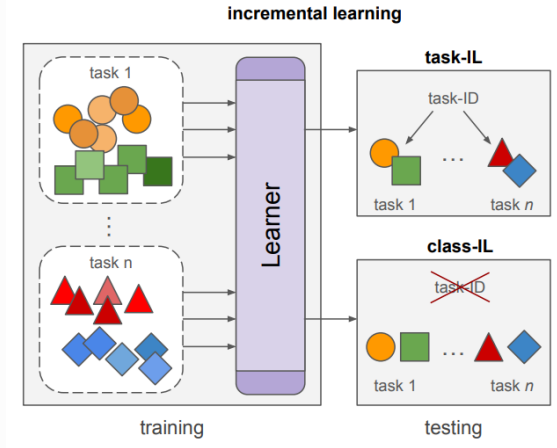
Left: image modified from [arxiv.org/pdf/2010.15277](https://arxiv.org/pdf/2010.15277) Right: image from [ieeexplore.ieee.org](https://ieeexplore.ieee.org) (arnumber=10444954)

# Class Incremental Learning



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# Class Incremental Learning



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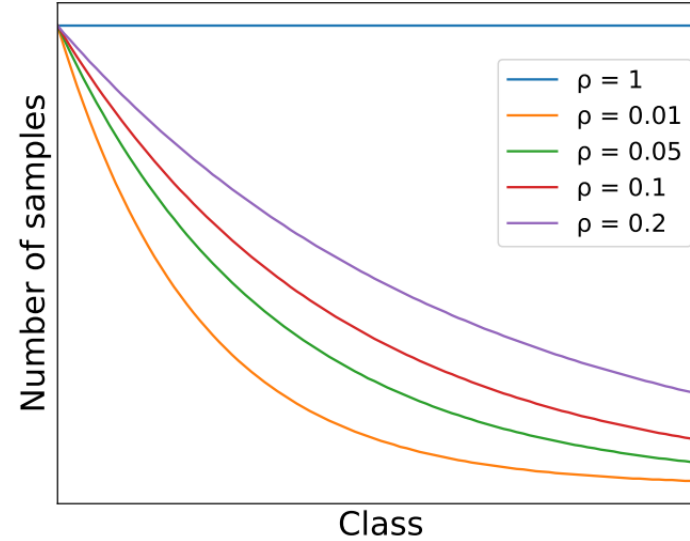


# Long-tailed learning

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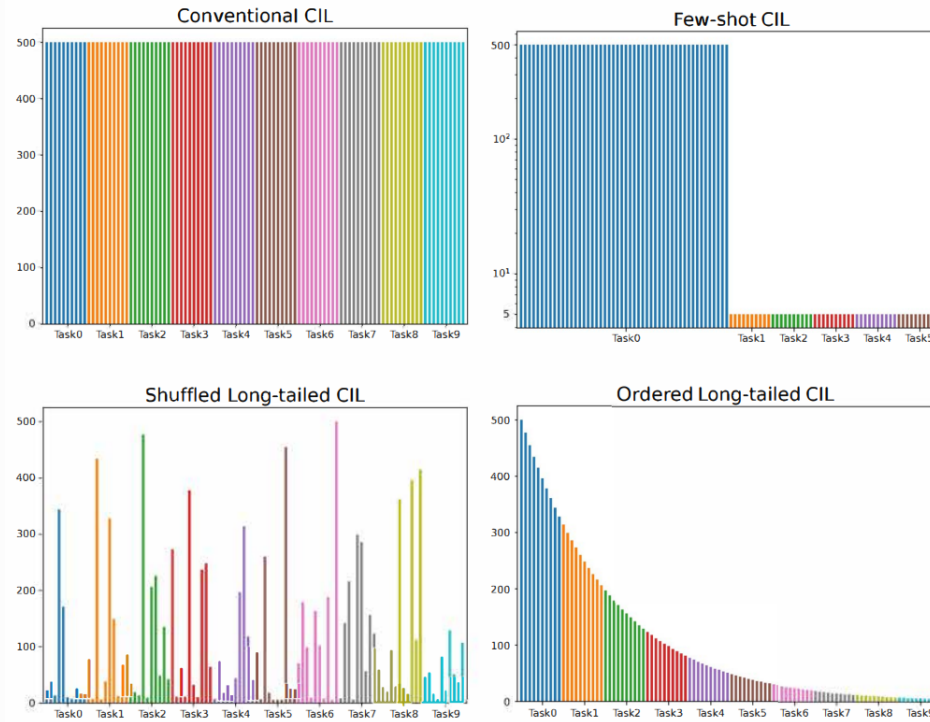
- re-sampling
- re-weighting
- transfer learning
- data augmentation

# Long-tailed learning



$\rho$  = ratio between most and least frequent classes

# LT-CIL scenarios



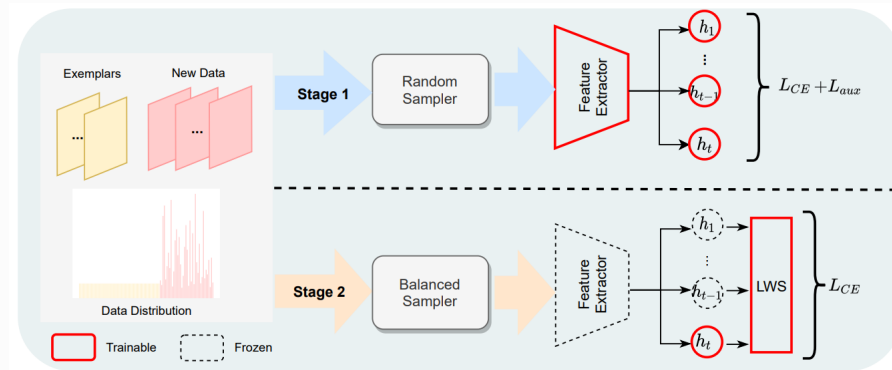
# Two-stage approach to LT-CIL

The classification models used in the study can be split into two parts:

- a **feature extractor**  $f_{\theta}$  with parameters  $\theta$
- a **classification head**  $h_{\phi}$  with parameters  $\phi$

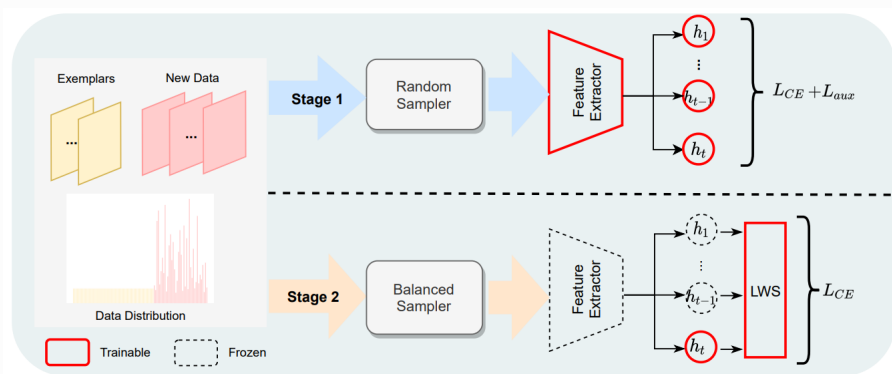
# Two-stage method for LT-CIL

1. Improve the feature extractor,  $f_\theta$ , using an *instance-balanced* sampler and  $\mathcal{L}_{CE,t} + \mathcal{L}_{aux,t}$



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1. Improve the feature extractor,  $f_\theta$ , using an *instance-balanced* sampler and  $\mathcal{L}_{CE,t} + \mathcal{L}_{aux,t}$
2. Retrain the classifier,  $h_\phi$ , using a *class-balanced* sampler, now with the **Learnable Weight Scaling (LWS)** layer, with  $\mathcal{L}_{CE,t}$





# Experimental setup

- **Datasets:** CIFAR-100 and ImageNet-Subset (100 classes)
- **Framework:** FACIL
- **Architectures:** ResNet-32 (CIFAR-100), ResNet-18 (ImageNet-Subset, iNaturalist)
- **Evaluation:** average accuracy over all classes and average incremental accuracy over all tasks
- **Imbalance ratio:**  $\rho = 0.01$
- **Exemplars per class:** 20

# Existing methods on LT-CIL scenarios

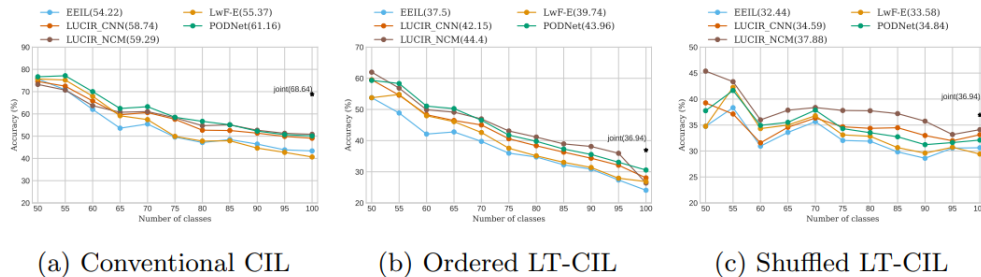


Fig. 4: Average accuracy for different scenarios on CIFAR-100. Average incremental accuracy is in the parentheses.

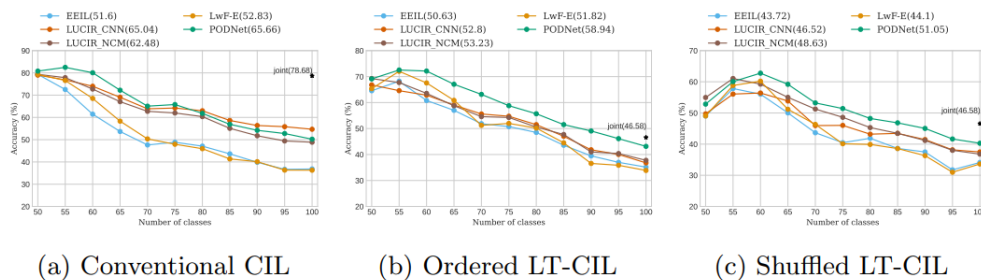


Fig. 5: Average accuracy for different scenarios on ImageNet-Subset. Average incremental accuracy is in parentheses.

# New method combined with existing methods

10 X. Liu et al.

| Methods                 |          | CIFAR-100         |                   | ImageNet-Subset   |                   |
|-------------------------|----------|-------------------|-------------------|-------------------|-------------------|
|                         |          | 5 tasks           | 10 tasks          | 5 tasks           | 10 tasks          |
| <i>Ordered LT-CIL</i>   | EEIL     | 38.46             | 37.50             | 50.68             | 50.63             |
|                         | + (Ours) | 38.97+0.51        | 37.58+0.08        | 51.36+0.68        | 50.74+0.11        |
|                         | LUCIR    | 42.69             | 42.15             | 52.91             | 52.80             |
|                         | + (Ours) | <b>45.88+3.19</b> | <b>45.73+3.58</b> | 54.22+1.31        | 55.41+2.61        |
|                         | PODNET   | 44.07             | 43.96             | 58.78             | 58.94             |
|                         | + (Ours) | 44.38+0.31        | 44.35+0.39        | <b>58.82+0.04</b> | <b>59.09+0.15</b> |
| <i>Shuffled LT-CIL</i>  | EEIL     | 31.91             | 32.44             | 42.87             | 43.72             |
|                         | + (Ours) | 34.19+2.28        | 33.70+1.26        | 49.31+6.44        | 48.26+4.54        |
|                         | LUCIR    | 35.09             | 34.59             | 45.80             | 46.52             |
|                         | + (Ours) | <b>39.40+4.31</b> | <b>39.00+4.41</b> | <b>52.08+6.28</b> | 51.91+5.39        |
|                         | PODNET   | 34.64             | 34.84             | 49.69             | 51.05             |
|                         | + (Ours) | 36.37+1.73        | 37.03+2.19        | 51.55+1.86        | <b>52.60+1.55</b> |
| <i>Conventional CIL</i> | EEIL     | 57.41             | 54.22             | 53.84             | 47.30             |
|                         | + (Ours) | 59.10+1.69        | 56.91+2.69        | 57.45+3.61        | 53.40+6.10        |
|                         | LUCIR    | 61.15             | 58.74             | 67.21             | 65.04             |
|                         | + (Ours) | 63.48+2.33        | 60.57+1.83        | 68.82+1.61        | 67.44+2.40        |
|                         | PODNET   | 63.15             | 61.16             | 70.13             | 65.66             |
|                         | + (Ours) | <b>64.58+1.43</b> | <b>62.63+1.47</b> | <b>71.08+0.95</b> | <b>68.47+2.81</b> |

Table 1: Comparison of average incremental accuracy on CIFAR-100 and ImageNet-Subset in the LT-CIL and conventional CIL scenarios.

# Ablation study on different imbalance ratios

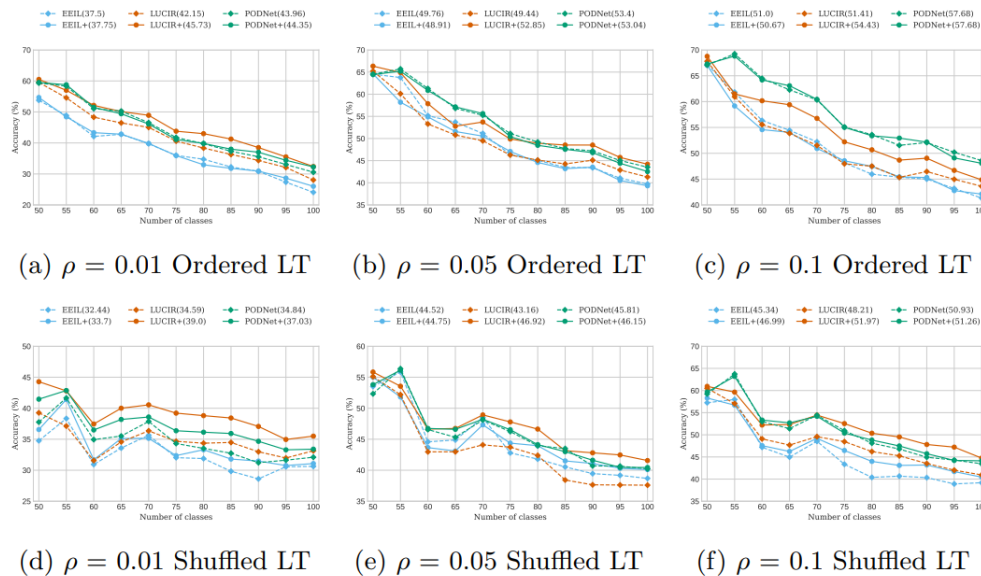
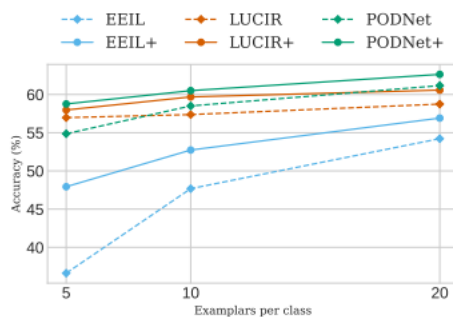
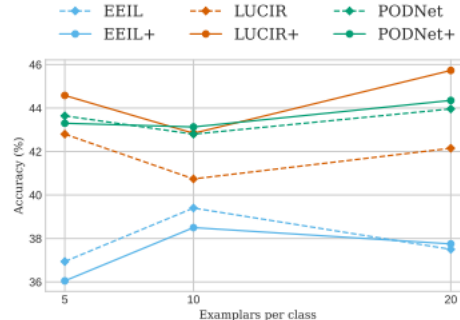


Fig. 6: Average accuracy on CIFAR-100 dataset for different imbalance ratios. The top row is for Ordered LT-CIL and the bottom row for Shuffled LT-CIL. The + suffix indicates our two-stage method applied to the corresponding baseline.

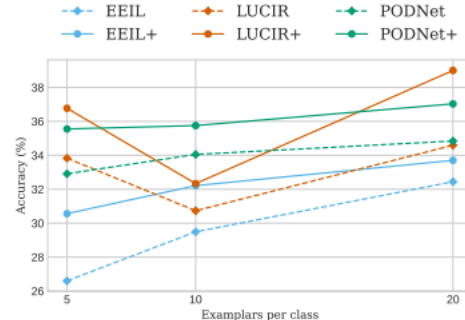
# Ablation study on impact of the number of stored exemplars



(a) Conventional CIL



(b) Ordered LT-CIL



(c) Shuffled LT-CIL

Fig. 7: Average incremental accuracy on CIFAR-100 for different scenarios as a function of stored exemplars. The + suffix indicates our two-stage method applied to the corresponding baseline.

# Ablation study on 25 task sequences

|              | EEIL  | EEIL+ | LUCIR | LUCIR+       | PODNet | PODNet+      |
|--------------|-------|-------|-------|--------------|--------|--------------|
| Conventional | 51.07 | 53.24 | 52.02 | 56.25        | 58.42  | <b>60.47</b> |
| Ordered      | 39.47 | 40.03 | 38.31 | <b>43.65</b> | 41.47  | 40.78        |
| Shuffled     | 32.20 | 34.68 | 30.68 | <b>37.38</b> | 35.75  | 37.09        |

Table 3: Average accuracy on long sequences of 25 tasks for three different scenarios on CIFAR-100. The + suffix indicates our two-stage method applied to the corresponding baseline.



# Key takeaways

- Proposes two novel scenarios for LT-CIL: **shuffled** and **ordered** LT-CIL



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- Proposes two novel scenarios for LT-CIL: **shuffled** and **ordered** LT-CIL
- Proposes a baseline two-stage method with a learnable weight scaling layer that compensates for class imbalance
- Improves several existing CIL methods in non-LT scenarios, possibly because limited memory buffer itself causes a data imbalance

